

Specifications (MiG-21F-13)

ENGINE 1 x 13,670lb (6,200kg) reheated Tumanskii turbojet

WINGSPAN 23ft 5in (7.2m)

LENGTH 51ft 9in (15.8m)

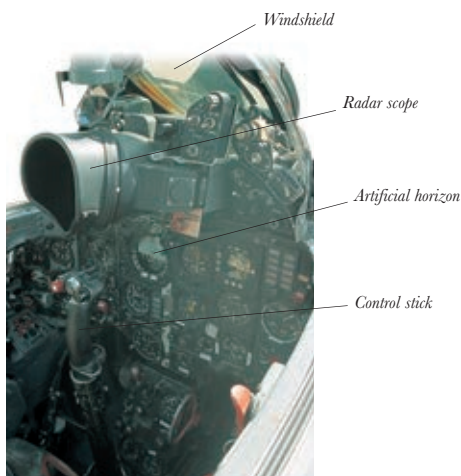
TOP SPEED 1,386mph (2,220kph)

CREW 1

ARMAMENT 1 x 30mm cannons; up to 4 x K-13 air-to-air missiles; 4 x 550lb (250kg) bombs or 4 x 220mm or 325mm air-to-surface missiles

FAST AND AGILE

Despite having only one engine, the MiG-21 is still an exceptionally agile aircraft that is not easily matched in dogfights. However, it is sometimes susceptible to snaking, which causes stalls and surges in the turbojet engine.



SIMPLE BUT EFFECTIVE

Keeping in line with the MiG-21's overall ethos, the cockpit is small and retains only the essentials in terms of flight and engine instruments.



STRIPPED DOWN

The MiG-21 (model 21F-13 shown here) is characterized by its relatively small size, delta-shaped wing configuration, and lightness. Keeping the weight to a minimum means that it can operate with just one engine.

Secretary of Defense Robert McNamara, one of the chief architects of the war, ended up believing that “the picture of the world’s greatest superpower killing or seriously injuring 1,000 noncombatants a week, while trying to pound a tiny backward nation into submission on an issue whose merits are hotly disputed, is not a pretty one.”

The reference to North Vietnam as “a backward nation” would have had an ironic ring to American airmen at the sharp end of Rolling Thunder. Flying from bases in South Vietnam and Thailand and from carriers of the Seventh Fleet in the Gulf of Tonkin, the fighter bombers had to contend with barrages of antiaircraft guns, Soviet-supplied SA-2 surface-to-air missiles (SAMs), and MiG-17 and MiG-21 fighters safely based in the “sanctuary” areas around Hanoi. This was state-of-the-art communist air-defense weaponry and it inevitably took its toll. The Americans lost a total of 938 aircraft in the three-year campaign, the majority to antiaircraft fire or SAMs. Fifty-six US aircraft are estimated to have been lost in air-to-air combat, compared with 118 MiGs downed by US fighters. Although this gave the Americans a comfortable 2-to-1 kill-ratio advantage, it was very different from the

experience over the Yalu River, when 10 MiGs had been shot down for every one American jet.

Strike force

The key aircraft in Rolling Thunder were the F-4 Phantom II as an air superiority fighter and the F-105 Thunderchief for ground attack. Neither was ideal for the tasks it was being asked to perform in Vietnam. The F-105 was a deep penetration fighter-bomber, intended to carry a nuclear bomb through Soviet air defenses. It was not designed for precision bombing of targets such as bridges and railroads. The F-4 was a powerful, versatile aircraft, but its designers had not envisaged it in traditional dogfights. It was meant to engage enemy aircraft with radar-guided Sparrow missiles beyond visual range, following up with heat-seeking Sidewinders when it got closer. It did not have a gun for an eyeball-to-eyeball confrontation.

The rules of engagement in Vietnam took away the Phantom II’s prime advantage from the outset, by insisting that an aircraft had to be visually identified before it was attacked. This meant that the MiG-17s and MiG-21s could get in close, where their maneuverability and their cannon gave them a fair chance. The North Vietnamese pilots consistently showed skill and aggression, although the Phantom pilots soon learned to exploit the